

ImpressionCore HCEP: Advanced Implementation Strategy

Leveraging Anthropometry & Head Pose for Gaze Inference

Version: 2.1 **Status:** Advanced Research & Engineering Draft **System:** ImpressionCore / HCEP Module

1. Executive Summary

The **Human Conversation Eye Points (HCEP)** system requires the avatar to understand not just *where* the user is, but *what* they are looking at (Logic/Left Eye, Emotion/Right Eye, Spiritual/Third Eye, Sincerity/Heart).

The Core Thesis: We do not need expensive iris-tracking hardware. By synthesizing **Head Pose Estimation (HPE)** from facial landmarks with **Anthropometric Constants** (average human head dimensions), we can calculate a **"Computed Gaze Vector."** This vector, when mathematically intersected with the avatar's virtual plane, provides the coordinates necessary to trigger HCEP responses.

2. Anthropometric Foundation (The "Base Data")

To accurately track a user using standard cameras, we must map the 2D image data to a known 3D physical model. We utilize biological averages to solve for depth and scale.

2.1 The Standard 3D Face Model

We establish a set of "World Coordinates" based on the average adult human face. These are our constants (in millimeters):

- **Inter-Pupillary Distance (IPD):** 63mm (Global Average).
- **Sellion (Bridge of Nose) to Chin:** 118mm.
- **Eye Depth (Offset):** The eyeball center is approx. 12mm posterior to the eyelid plane.
- **The "Cyclopean Eye":** The theoretical center point between the two eyes (Bridge of nose), which serves as the origin point for our primary gaze ray.

Implementation Rule: When the system detects a face, it scales the observed 2D landmarks to match these 3D constants. If the user's IPD measures 63 pixels on screen, and we know the average is 63mm, we can calculate the exact Distance (Z-depth) from the camera.

3. Engineering The Gaze Vector (The Math)

The transition from "Face Tracking" to "HCEP Gaze Tracking" involves three mathematical stages.

Stage 1: SolvePnP (Head Pose Estimation)

We use the **Perspective-n-Point (PnP)** algorithm.

- **Input:** 2D Image Landmarks (Eye corners, Nose tip, Chin, Ears).
- **Reference:** Our 3D Anthropometric Model.
- **Camera Matrix:** Focal length and optical center (intrinsic parameters).
- **Output: Rotation Matrix (R) and Translation Vector (t).**

This tells us exactly how the user's head is tilted (Yaw, Pitch, Roll) and where it is in 3D space relative to the camera.

Stage 2: The "Hybrid Gaze" Calculation

Head orientation alone is not enough; eyes can move independently. We implement a **Hybrid Weighted Model**:

$$\vec{V}_{gaze} = (w_h \cdot \vec{V}_{head}) + (w_e \cdot \vec{V}_{pupil})$$

1. **Head Vector ($\vec{V}_{head}$):** The forward-facing vector derived from the Rotation Matrix.
2. **Pupil Vector ($\vec{V}_{pupil}$):** The offset of the iris center relative to the eye corners (using facial recognition landmarks).
3. **Weights (w):**
 - If the user's head is moving fast, w_h (Head Weight) increases (eyes usually lock forward during head turns).
 - If the head is still, w_e (Eye Weight) increases (user is scanning with eyes only).

Stage 3: Ray-Plane Intersection (The HCEP Trigger)

We treat the Avatar's display as a **Virtual Plane** in 3D space.

1. We cast a ray starting from the user's **Cyclopean Eye** (User's location).
2. We project this ray along the $\vec{V}_{gaze}$.
3. We calculate the intersection point (x, y) on the Screen Plane.

The Formula:

$$P_{screen} = P_{user} + d \cdot \vec{V}_{gaze}$$

Where d is the scalar distance to the screen plane.

4. HCEP Zone Mapping & Logic

Once we have the intersection point (x, y) on the screen (e.g., "The user is looking at pixel 900, 400"), we map this to the Avatar's anatomy.

4.1 The Zones (Target Areas)

Zone	Anatomy	Emotional Context	Data Signal
Zone A	Left Eye	Logic & Analysis	User is fact-checking, analyzing syntax, or looking for structural errors.
Zone B	Right Eye	Emotion & Empathy	User is seeking emotional validation, sincerity, or subtext.
Zone C	Glabella (Third Eye)	Spiritual / Meta	User is accessing intuition or judging the "soul"/intent of the avatar.
Zone D	Heart / Chest	Sincerity / Trust	User is "grounding" the conversation or checking for non-verbal honesty.

4.2 The "Confidence Cone"

Because webcam tracking has jitter, we do not use single pixels. We use **Probability Cones**.

- If the Gaze Ray lands within 5cm of the Left Eye, `Logic_Score` increases.
- We use a **Dwell Time Filter**: The gaze must hold for >200ms to register as an HCEP signal, filtering out random saccades.

5. Implementation into ImpressionCore

Step 1: Input Layer Update

Update the current `InputManager` to accept dual streams:

1. **Depth Stream (if available)**: Kinect/Realsense (Preferred for Z-depth).
2. **RGB Stream**: Standard Camera.

Step 2: The "FaceEngine" Class

Create a new class in `ImpressionCore`:

```

class FaceEngine:
    def process_frame(self, image):
        # 1. Detect Landmarks (MediaPipe/Dlib)
        landmarks = detect_mesh(image)

        # 2. Solve PnP to get Head Rotation (R) and Translation (t)
        R, t = solve_pnp(landmarks, model_points_3d, camera_matrix)

        # 3. Calculate "Cyclopean" Origin
        origin = (landmarks.left_eye + landmarks.right_eye) / 2

        # 4. Raycast to Screen Plane
        gaze_point = intersect_ray_plane(origin, R, screen_plane)

        return gaze_point

```

Step 3: Integration with Avatar Controller

The Avatar Controller currently tracks the user's position to rotate the head. We add a secondary layer: **Eye Contact Logic**.

- **Current:** Avatar Head rotates to face User ($U_{ser_{pos}}$).
- **New:** Avatar Eyes rotate to face the *User's Eyes*.
- **New:** If `gaze_point` hits Avatar's Left Eye -> Avatar triggers `MicroExpression_Analytical`.
- **New:** If `gaze_point` hits Avatar's Right Eye -> Avatar triggers `MicroExpression_Empathetic`.

6. Advanced Behavior: Reciprocal Gaze

The "Holy Grail" of HCEP is not just knowing where the user looks, but having the Avatar **mirror** natural human behavior.

6.1 The Saccade Triangle

When the Avatar speaks, it should not stare fixedly. It should trace the **Social Triangle**:

1. Look at User's Left Eye (0.5s)
2. Look at User's Right Eye (0.5s)
3. Look at User's Mouth (while User is speaking) or Third Eye (while Avatar is thinking).

6.2 Mutual Gaze Lock

If the system detects the user looking at the Avatar's Right Eye (Emotion), the Avatar should:

1. **Slow Blink.**
2. **Dilate Pupils** (if rendering allows).
3. **Soften Facial Muscles** (Action Units 6 & 12 - Orbicularis Oculi). This creates a bio-feedback loop of empathy.

7. Conclusion

By relying on the **Geometry of the Skull** and **Eye-Head Synergy**, ImpressionCore can achieve